Database Backup and Recovery

backup and recovery refers to the various strategies and procedures involved in protecting your database against data loss and reconstructing the database after any kind of data loss

Database Backup

- Backup involves making copies of your database to recreate the database if necessary.
- Backups can be divided into **physical backups** and **logical backups**.
- **Physical backups** are backups of the physical files used in storing and recovering your database, such as data files, control files, and archived redo logs.
- **Logical backups** contain logical data (for example, tables or stored procedures) exported from a database with an Oracle export utility and stored in a binary file.
- Logical Backups are an adjunct, not an alternative, to physical backups.
- Backups can be performed in two different ways:
  - Server managed backup (using Oracle Recovery Manager –RMAN)
  - User managed backup (Using operating system utilities, data pump export utility)

Database Backup…

- Backup may be:
  - Cold backup (closed backup or consistent backup or offline backup)
    - Backup taken when the db is shutdown
    - Doesn’t need to go through recovery process
  - Hot backup (open backup or inconsistent backup or online backup)
    - Backup carried out when db is in use
    - Can only be made if the db is in archivelog mode
    - Always needs to undergo a recovery after restore
  - A full backup includes all used blocks of the files backed up
  - An incremental backup includes only those blocks that have been changed since the last backup
    - An incremental backup can be cumulative (including all blocks changed since the last full backup) or differential (including all blocks changed since the last incremental backup)
  - Whole and Partial database backup
    - You can backup either entire database or part of it such as tablespace or datafile
    - Partial backups are only possible if database is in archivelog mode

Database recovery

- Reconstructing the contents of all or part of a database from a backup typically involves two phases: retrieving a copy of the data file from a backup, and reapplying changes to the file since the backup from on the archived and online redo logs, to bring the database to the desired SCN (usually the most recent one)
- Restoring database involves to retrieve the data files or control file from a backup location on tape, disk or other media, and make it available to the database server.
- Recovering the database means taking a restored copy of the data files and apply to them changes recorded in the database's redo logs.

Steps in restore and recovery

- Restore
  - Extract the datafile and control file from pervious backup made.
  - Using backed-up copies of data files and control files to replace the lost or damaged data files and control file is called restoring
- Recover
  - Apply changes extracted from redo logs to keep the database up to date
  - Bringing the datafile up to date using backed up data files and redo log files is called recovery
- Oracle Recovery process
  - Crash and instance recovery
    - Automatically done when instance suddenly fails
    - Instance recovery is very similar to crash recovery, but it applies to cases where a surviving instance recovers the failed instance in an RAC (Real Application clustering) env.
    - Crash recovery is in single instance env
    - Uses current data files and apply online redolog files only for the recovery
    - Don't apply backed up datafiles, archive redo logs during recovery
    - This involves 2 steps
      - Rolling Forward (Cache recovery): applies committed and uncommitted data in the current online redo log files to current online data files.
      - Rolling back (transaction Recovery): removes the uncommitted transactions applied in the previous step, using the undo data in the undo segments.

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- Media Recovery
  - It is not automatic; the DBA has to initiate the recovery process
  - This is done if the data files are lost or damaged
  - You need following files to perform a complete media recovery
    - A full backup of all datafiles
    - Archive redo logs since the last full backup
    - A copy of control file
    - Current online redo log files
  - This involves 2 steps
    - Restore
      - » Restore the backup of the data files and make them available to oracle
    - Recover
      - » Then bring the data files up-to-date by applying the archive redo logs and online redo logs
      - » The recovery process itself has 2 steps
        - Cache recovery (rolling forward)
          - Apply both committed and uncommitted data from archive redolog and online redolog to the data fiels
        - Transaction Recovery (rolling back)
          - The uncommitted changes are removed from datafiles using undo segments

Backup recovery Guideline

- Build redundancy into your system by using RAID based storage system which will let you mask individual disk failure.
- Perform backup at frequent intervals to reduce your recovery time.
- Maintain offsite storage of your backups with a reliable vendor.
- Keep database in archive log mode
- Multiplex the control files, online redo log files and archive redo logs
- After every major structural changes backup the control files.
- Always make more than one copy of backups
- Backup other files like init.ora and server parameter file (spfile.ora), network related file (tnsnames.ora, listner.ora etc), password file etc to reduce the downtime
- Use RMAN form backup and recovery as RMAN maintains a log of all backup and recovery actions performed, so its easy to keep track of those operations.
- Use the datapump export utility (logical backup) to provide supplement protection
- Maintain tha DR site to replicate the database at remote end
- Maintain a redundancy set:

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- Maintain a redundancy set:
  - Always keep redundancy set online so you can recover faster
  - A redundancy set is defined as:
    - Last backup of all datafiles
    - Last backup of control file
    - Multiplexed copies of current redolog files
    - Copies of current control file that being used
    - All the archive redo logs since the last backup
    - Other files like spfiles.ora or init.ora, listner.ora, tnsnames.ora etc
  - Make sure that you have the redundancy set on completely separate physical volume and RAID system than those on which the datafiles, online redolog files, control file are located to ensure no loss of data incase of media failure.

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- Test the backup and recovery periodically.
  - To avoid a catastrophic recovery experience, every DBA should have established and tested backup and disaster recovery plan.
- Establish a recovery policy
  - Schedule the db recovery in UAT env
  - DR drill
- Always validate your backup and make sure that backups are actually readable
    - Try to restore the backup and recover the database time to time to increase your level of confidence.